

Offline, Confidence-Aware Flood Routing with Verified Explanations: Design and a Replay Evaluation on the Chennai Road Network

(Working title)

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Abstract—Navigation apps route drivers into flooded streets because they optimise travel time on a network assumed to be intact, and they often stop working when floods take down cellular service. We describe CityPulse AI, a routing system for Chennai, India, that fuses a static flood-susceptibility prior with timestamped, source-weighted hazard reports; routes on a pessimistic upper bound of the fused hazard probability whose caution grows as evidence thins; explains each route through a deterministic template or a language-model rewrite that must pass a symbolic check against the router’s decision record; and runs entirely on the phone. We evaluate the router on the OpenStreetMap road graph of Chennai (193,191 nodes, 471,240 directed edges) with a replay corpus of 6,132 geolocated observations from the 2015 floods. Scoring every configuration under one common reference belief, rather than under each configuration’s own belief as in our earlier harness, changes the conclusions: the default commuter setting altered 7 of 100 routes and left 29% of routes crossing an edge with fused probability of at least 0.5, against 30% for hazard-blind routing. With risk weights between 5 and 20, the expected number of hazardous edges per route fell from 1.61 to between 0.98 and 0.18, at a mean travel-time cost of 0.85% to 12.7%. A calibration study against a source-holdout label found weak discrimination (area under the ROC curve 0.61) and no advantage of class-specific decay over simpler baselines. The corpus has a single timestamp, no depth measurements and no negative reports, so these results characterise the method on historical data and do not establish safety in live use.

Index Terms—Explainable artificial intelligence, flood hazards, mobile computing, route planning, uncertainty.

I. INTRODUCTION

Urban floods disrupt road networks long before water becomes visible on a map. Hydrodynamic studies show that pluvial floods disconnect intra-urban networks [1], [2] and stretch emergency response times [3], [4], and standing water slows traffic in a way that can be quantified through depth-disruption functions [5]. Chennai experienced major floods in 2015 and 2023 [6], [7], and urban growth is expected to increase exposure [8].

Three shortcomings of current navigation matter here. First, hazard information does not reach the router in a usable

form: reports from residents and agencies are sparse, stale and of uneven reliability [9], [10]. Second, routes come without reasons, and generated reasons risk inventing facts [11]. Third, cloud-dependent services fail when connectivity does, which is common during floods.

Flood-aware routing itself is not new; it appears in recent studies [12]–[14] and in patents [15], [16]. What we examine is a specific combination: routing on a belief that fuses a prior with decaying crowd evidence, becoming more cautious when evidence is thin, with explanations that are verified against the decision record, all on the device. This paper makes the following contributions:

- A formulation that separates slowdown from harm, uses a pessimistic upper bound of the hazard probability with a per-user-class caution parameter, and an argument that confidence-multiplied penalties are risk-seeking under ambiguity (Section III).
- A bound showing that a time-proportional hazard penalty can raise an edge’s cost by at most a factor of $1 + \lambda s$, which explains why the default commuter setting barely changes routes (Proposition 2 and Section VI).
- The design of an offline-first system with a structured decision trace and a fail-closed explanation verifier (Section IV).
- A replay evaluation on the full Chennai road graph that scores all configurations under a common reference belief, reproduces the production router’s paths with an independent implementation, and reports a calibration study whose results are weak (Sections V–VI).

II. RELATED WORK

A. Floods, road networks and flood-aware routing

Coupled flood and network models quantify accessibility loss [1]–[4], [17], and depth-disruption functions relate water depth to safe speed [5], [18]. Flood-aware route planning has been built on numerical flood models with Dijkstra’s algorithm [12], combined with rescue dispatch [13], learned with

reinforcement learning [14], and applied to evacuation [19], [20]. These studies generally treat the flood state as known from a model. We instead treat it as an uncertain belief built from sparse reports.

B. Routing under uncertainty and changing weights

Stochastic and reliable shortest paths optimise expected time or on-time arrival [21]–[25]; chance constraints [26] and robust optimisation [27] express caution. For fast queries under changing weights, customisable hierarchies separate preprocessing from customisation [28], [29], and landmark (ALT) search [30] remains correct as long as no edge weight drops below its initial value [31]. Our cost satisfies this condition by construction; we use the property and credit it to that work.

C. Crowd evidence, belief and calibration

Crowdsourced and navigation-app reports can indicate flooded roads [10], [32], [33], and truth-discovery methods estimate source reliability [34], [35]. Decay of trust with age is established in reputation systems [36], [37], and the persistence filter models whether a mapped feature still exists [38]. We evaluate probabilities with the Brier score and its decomposition [39], [40] and report discrimination separately [41].

D. Explanation and on-device generation

Explanations should be contrastive [42] and faithful [43]. Template generation is well suited to closed domains [44], [45], while neural generation hallucinates [11] and is checked statistically in open domains [46], [47]. Small quantised models now run on phones [48], [49], but measured performance varies widely across devices [50], [51]. Offline-first designs rely on logical clocks and replicated data types [52]–[54].

III. PROBLEM FORMULATION

A. Road graph and hazard belief

Let $G = (V, E)$ be a directed road graph with free-flow traversal time $\tau_0(e) > 0$. For edge e and hazard class c , a static prior log-odds $\ell_0(e)$ encodes susceptibility. An observation i has location x_i , time t_i , polarity $y_i \in \{+1, -1\}$ and source reliability $\alpha_i \in (0, 1)$. With spatial kernel $\kappa(\cdot)$ and class time constant T_c , the fused belief at query time t is

$$\ell(e, t) = \ell_0(e) + \sum_i y_i \omega_i(e, t) \logit(\alpha_i), \quad (1)$$

$$\begin{aligned} \omega_i(e, t) &= \kappa(d(e, x_i)) e^{-(t-t_i)/T_c}, \\ \bar{p}(e, t) &= \sigma(\ell(e, t)), \quad n_{\text{eff}}(e, t) = \sum_i \omega_i(e, t). \end{aligned} \quad (2)$$

The additive log-odds form follows the occupancy-grid tradition [55]; it assumes that observations are conditionally independent given the hazard state, an assumption we revisit in Section VII.

B. Pessimistic plug-in

The router uses the upper bound

$$\tilde{p}(e, t) = \min \left\{ 1, \bar{p} + z \sqrt{\bar{p}(1 - \bar{p}) / (n_{\text{eff}} + 1)} \right\}, \quad (3)$$

where $z \geq 0$ is set per user class (for example $z = 0$ for commuters and $z = 2$ for emergency vehicles). For fixed $\bar{p} \in (0, 1)$ and $z > 0$, $\tilde{p}$ decreases as n_{eff} grows: thin evidence produces more caution, not less. This mirrors the upper confidence bounds of bandit algorithms [56] but is used pessimistically.

Why not scale the penalty by confidence? A common alternative prices an edge as $w = \tau_0(1 + \lambda RC)$, with R a risk score and $C \in [0, 1]$ a confidence that decays with report age. As $C \rightarrow 0$ the penalty vanishes, so an unverified or ageing flood report makes the road look clear. For a vehicle that must avoid water, this is risk-seeking under ambiguity. Equation (3) reverses the direction of that dependence.

C. Edge cost and chance constraint

Slowdown and harm enter separately:

$$w_\lambda(e, t) = \tau_0(e) [1 + \tilde{p}(\delta - 1)] + \lambda \tilde{p} s_c \tau_0(e), \quad (4)$$

where $\delta = \max\{1, v_0(e)/v_{\text{safe}}(h)\}$ is the slowdown implied by a depth-disruption function [5] at depth h (and $\delta = 1$ when depth is unknown), s_c is the severity of class c , and $\lambda \geq 0$ is the risk weight. An edge is removed when a reported depth exceeds the class threshold h_{max} and $\tilde{p} \geq \varepsilon$, a point approximation of $\Pr[h > h_{\text{max}}] \leq \varepsilon$ [26].

Proposition 1 (Consistency of free-flow landmarks). *If $\delta \geq 1$, $\tilde{p} \geq 0$ and $\lambda \geq 0$, then $w_\lambda(e, t) \geq \tau_0(e)$ for every edge. Landmark potentials computed once on τ_0 therefore remain feasible, and ALT queries remain exact, under every hazard configuration.*

Proof: Both terms added to τ_0 in (4) are non-negative, so no weight drops below its free-flow value; correctness of ALT under such changes is shown in [31]. ■

The condition $\delta \geq 1$ does not follow from the depth-disruption function alone; for shallow water on a slow street the ratio v_0/v_{safe} is below one, so the implementation must clamp it.

Proposition 2 (Bounded deterrence of a time-proportional penalty). *When depth is unknown ($\delta = 1$), $w_\lambda(e, t)/\tau_0(e) = 1 + \lambda \tilde{p} s_c \leq 1 + \lambda s_c$. A route that avoids a set of hazardous edges with total free-flow time T_h is preferred only if its extra free-flow time is below $\lambda \sum_e \tilde{p}_e s_c \tau_0(e) \leq \lambda s_c T_h$.*

Proof: Substitute $\delta = 1$ in (4) and use $\tilde{p} \leq 1$. The comparison follows by summing the costs of the two routes. ■

For the commuter setting $\lambda = 0.3$ and $s_c = 1$, a 60 s flooded segment justifies at most an 18 s detour even when $\tilde{p} = 1$. Short, severe segments such as underpasses are cheap to traverse under this penalty. Section VI shows the consequence on real data.

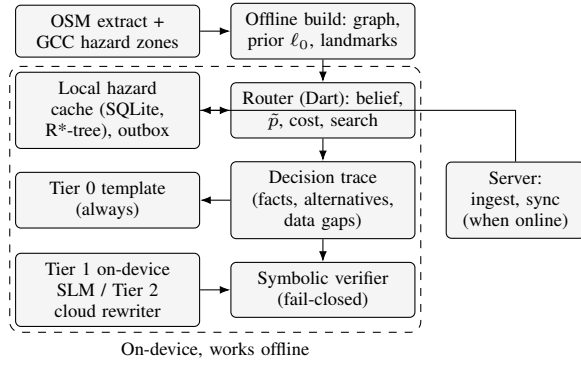


Fig. 1. CityPulse architecture. Everything inside the dashed box runs on the phone without connectivity.

IV. SYSTEM DESIGN

Fig. 1 shows the architecture. A single Dart routing package (compressed-sparse-row graph, bidirectional Dijkstra with ALT landmarks) is imported by the Flutter client and compiled to a command-line tool for the evaluation harness, so that the evaluated router and the shipped router are the same code. A belief package implements (1)–(3) as pure functions and rejects malformed inputs (for example future-dated observations or out-of-range reliabilities) with runtime errors rather than assertions, because a not-a-number value would otherwise make the chance constraint fail open. The client stores observations in SQLite with an R*-tree index [57], queues local reports in an outbox, and synchronises through an append-only log ordered by hybrid logical clocks [52], which behaves as a grow-only set replicated data type [53].

A. Decision trace and explanation tiers

Every query returns a decision trace containing the chosen route (duration, distance, free-flow duration, hazard time penalty, worst edge probability), the alternatives the router rejected with their blocking edges (hazard class, $\bar{p}$, $\tilde{p}$, n_{eff} , age of the newest observation, time penalty, depth if known), contextual facts, and explicit data gaps. Explanation layers may read only the trace. Tier 0 renders it with templates and needs no model. Tiers 1 and 2 ask a small on-device model or a cloud model to rewrite the facts; the rewrite is accepted only if it passes the verifier within a 1.2 s deadline, and otherwise the Tier 0 text is shown without any indication that a rewrite failed.

B. Verifier

Table I lists the checks as implemented. They are symbolic and deterministic, which is possible because the domain is closed. They are also deliberately incomplete: the numeral check does not bind a number to its unit or subject, and the entity check relies on capitalised words, which does not transfer to Tamil script. We have not yet measured the pass rate of model rewrites, so this paper reports the design only.

TABLE I
EXPLANATION VERIFIER CHECKS AS IMPLEMENTED

Check	Rejects a candidate text if ...
Numerals	a number is not within a declared tolerance of a trace value or its unit conversion (s/min, m/km, probability/%)
Entities	a capitalised word is not in a trace street name, corridor, label or route identifier; a hazard noun names a class absent from the trace
Comparatives	a speed comparison is unsupported, or an avoidance claim names an alternative without blocking edges
Data gaps	a sentence attaches a number, hazard noun or avoidance claim to a corridor listed as having no data
Hedging	the trace confidence is low or stale and no hedge phrase is present
Absolute safety	the text asserts passability or safety (denylist)

V. EXPERIMENTAL SETUP

A. Data

Graph. We built a drivable graph of Chennai from OpenStreetMap [58] with 193,191 nodes and 471,240 directed edges; free-flow speeds are planning defaults by road class. *Prior.* 7,453 flood-hazard zone polygons from the Greater Chennai Corporation, published through OpenCity, were mapped to edges within 600 m, with category-to-probability defaults from 0.02 (very low) to 0.45 (very high). These defaults are not calibrated, and no elevation or HAND [59] layer was available. *Corpus.* 6,132 geolocated observations from three OpenCity 2015 Chennai layers (5,379 flood, 753 waterlogging; 1,080 official, 5,052 crowd) were snapped to edges (median 2.9 m, 90th percentile 23.6 m). The source files carry no per-point dates, so every observation is stamped 2015-12-02T00:00Z; none carries depth; all are positive reports. *Hazard configuration.* 14,534 edges received a hazard entry: 5,775 with observations and 8,759 with prior probability at least 0.25. On 124 edges with two hazard classes, the minority class was dropped because the router accepts one class per edge.

B. Configurations

We compare C0, free-flow routing; C1, hard avoidance that removes edges with $\bar{p} \geq 0.5$; and C3, the method of Section III with class-specific decay, swept over $z \in \{0, 0.5, 1, 1.28, 2\}$ and $\lambda \in \{0.3, 1, 2, 5, 10, 20\}$. Configurations that differ only in decay (single decay constant, fixed time-to-live, and a hindsight oracle) produce identical beliefs on a corpus with one timestamp, so we report them only in the calibration study.

C. Protocol and metrics

We used the 100 origin–destination pairs sampled with the harness’s pinned seed. Our earlier harness scored each configuration with the worst $\tilde{p}$ on its own chosen route, computed under its own hazard set and its own z , and measured detours with the router’s objective, which includes penalty seconds. Both choices make configurations incomparable. Here every chosen route is scored under one reference belief, the C3

fused $\bar{p}$ at the replay clock (independent of z), and detours are measured in free-flow time. Metrics are the expected number of hazardous edges per route $\sum_{e \in P} \bar{p}_e$, the share of routes containing an edge with $\bar{p} \geq 0.5$, the free-flow-time-weighted mean $\bar{p}$, and the extra free-flow time relative to C0. Paired differences carry 95% bootstrap intervals (5,000 resamples, seed 20261001).

The C0, C1 and C3 ($z = 0$, $\lambda = 0.3$) routes are the production router’s own traces. For the sweep we re-implemented (4) with SciPy’s Dijkstra on the same graph and hazard entries. It reproduced the production router’s C3 node sequences on 100 of 100 pairs. The command-line router costs about 3.07 s per call on our development machine, almost all of it graph loading, which is why the harness used 100 rather than 1,000 pairs.

D. Calibration study

With no confirmed-absent reports, we used a source-holdout proxy: an edge is labelled positive if it has an official observation, and predictors see only crowd reports and the prior. Report ages of 0, 1, 6, 24, 72 and 168 h were simulated from the single timestamp, giving 34,256 edge–age rows with 708 positives. Baselines are a single shared decay constant (C2), a fixed six-hour time-to-live (C4), and a beta reputation model with forgetting [36].

VI. RESULTS

A. The default commuter setting is nearly hazard-blind

Under the common reference, C3 with $z = 0$ and $\lambda = 0.3$ changed 7 of 100 routes, added 0.026 s of free-flow time on average (95% CI [0.003, 0.061]), and reduced the expected number of hazardous edges per route by -0.085 ($[-0.250, 0.002]$); 29% of its routes still crossed an edge with $\bar{p} \geq 0.5$, compared with 30% for C0 (Table II). This is the behaviour Proposition 2 predicts: with $\lambda = 0.3$ a hazardous edge costs at most 1.3 times its free-flow time. Hard avoidance (C1) changed 30 routes, removed all crossings of $\bar{p} \geq 0.5$ edges (by construction, because its threshold equals the metric’s), and cost 27.3 s on average ([16.9, 38.9]), a 1.53% mean detour with a 90th percentile of 6.2%.

B. Larger risk weights trade time for exposure smoothly

Fig. 2 and Table II show the sweep. Increasing λ lowered exposure steadily: at $z = 0$, $\lambda = 5$ reduced the expected number of hazardous edges from 1.61 to 0.98 at a 0.85% mean detour, and $\lambda = 20$ reduced it to 0.44 at 4.53%. Against hard avoidance, the soft penalty reached lower expected exposure at a smaller mean detour (0.98 at 0.85% versus 1.22 at 1.53%), but left more routes touching a $\bar{p} \geq 0.5$ edge (21% versus 0%). Which is preferable depends on whether harm is driven by cumulative exposure or by the single worst segment.

C. Pessimism cannot be credited on this corpus

At matched detour, higher z did not lower reference exposure; the $z = 0$ curve lies on or below the others in Fig. 2. This is expected rather than informative. The pessimistic bound

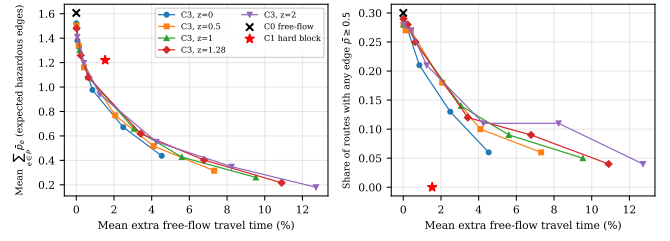


Fig. 2. Exposure against mean extra free-flow time for the C3 sweep ($\lambda \in \{0.3, 1, 2, 5, 10, 20\}$ along each line), free-flow routing (C0) and hard avoidance (C1), all scored under the same reference belief. In the right panel C1 is zero by construction because its blocking threshold equals the metric threshold.

shifts avoidance toward edges whose belief is uncertain, and its benefit can appear only against ground truth that the belief might be wrong about. Scoring against the belief itself cannot reward it. Our earlier harness reported the share of routes with worst $\tilde{p} > 0.5$ rising from 36% at $z = 0$ to 72% at $z = 2$; that increase came from measuring each configuration with its own inflated $\tilde{p}$, not from riskier routes. Similarly, the earlier mean detour of 0.27% for C3 counted penalty seconds as travel time; in free-flow time it was 0.0013%.

D. Calibration

Table III summarises the calibration study. All exponential-decay variants reached similar Brier scores and an area under the ROC curve of 0.609, and the class-specific decay (C3) was indistinguishable from a single decay constant (C2). Resolution was almost zero (6.9×10^{-5} against an uncertainty of 0.0202), and the reliability diagrams (Fig. 3) show predicted probabilities far above observed frequencies. The beta reputation baseline scored worst, largely because it falls back to the prior; the comparison is therefore not a test of forgetting as such. On the crowd-only pool, C3 reached an area under the curve of 0.566. Because the label is a proxy and ages are simulated, we do not read these numbers as evidence for or against class-specific decay.

E. Engineering checks

Two runs of the replay harness on 50 pairs produced byte-identical trace files (matching SHA-256 digests). Over the whole sweep, the minimum ratio w_λ/τ_0 was exactly 1.0, as Proposition 1 requires. Among edges with observations, the median fused $\bar{p}$ was 0.137 and 6.3% had $\bar{p} \geq 0.5$; prior-only edges had a median of 0.286. The repository reports about 250 automated tests across the routing, belief and explanation packages, the client and the server.

VII. THREATS TO VALIDITY

- *Circular reference.* Exposure is measured under the model’s own fused belief. It shows how configurations trade time for believed risk, not whether they avoid real flooding. Ground truth independent of the belief is required.

TABLE II
ROUTE EXPOSURE UNDER A COMMON REFERENCE BELIEF (C3 $\bar{p}$ AT THE REPLAY CLOCK), 100 ORIGIN–DESTINATION PAIRS, 2015 CHENNAI CORPUS

Config.	z	λ	Mean extra time (%)	P90 extra time (%)	$\sum \bar{p}$ per route	Share $\bar{p} \geq 0.5$ (%)	TW-mean $\bar{p}$
C0 (free-flow)	—	—	0.00	0.00	1.61	30	0.0113
C1 (block $\bar{p} \geq 0.5$)	0	0	1.53	6.20	1.22	0	0.0078
C3	0	0.3	0.00	0.00	1.52	29	0.0100
C3	0	5	0.85	2.73	0.98	21	0.0062
C3	0	20	4.53	13.93	0.44	6	0.0023
C3	1	5	3.06	10.51	0.66	14	0.0037
C3	1	20	9.53	25.57	0.26	5	0.0009
C3	2	0.3	0.05	0.07	1.41	28	0.0094
C3	2	5	4.25	12.20	0.55	11	0.0028
C3	2	20	12.72	30.97	0.18	4	0.0005

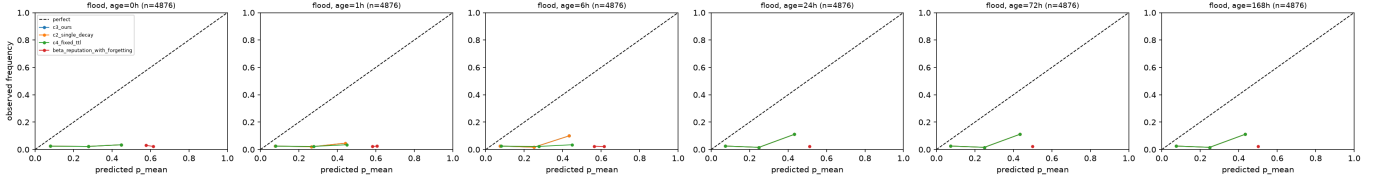


Fig. 3. Reliability diagrams for the flood class by simulated report age (0 to 168 h), from the project’s calibration study. All models over-predict relative to the proxy label.

TABLE III
CALIBRATION AGAINST THE SOURCE-HOLDOUT LABEL (34,256 ROWS, 708 POSITIVES)

Model	Brier	Reliab.	Adapt. ECE	Log loss	AUROC
C3 class-specific decay	0.0343	0.0137	0.082	0.159	0.609
C2 single decay	0.0344	0.0137	0.082	0.159	0.609
C4 fixed TTL (6 h)	0.0367	0.0161	0.089	0.167	0.609
Beta reputation [36]	0.2921	0.2712	0.520	0.779	0.568

- *Corpus*. One event, one timestamp, no depth and no negative reports: decay, the chance constraint, false-avoidance cost and event-split validation cannot be tested.
- *Independence*. Equation (1) treats reports as independent. Crowd reports of one flood are correlated (2,806 duplicates by map identifier were already removed upstream), which inflates certainty.
- *Priors and speeds*. Category probabilities and free-flow speeds are defaults, not fitted values.
- *Scale*. One hundred pairs, 30 sweep points, one city.
- *Explanation*. The verifier is described, not evaluated; no on-device timing on a low-cost phone is reported.

VIII. CONCLUSION AND FUTURE WORK

We asked whether a router that fuses a flood prior with decaying reports, and grows more cautious as evidence thins, changes routes in a useful way on a real city graph. On the 2015 Chennai corpus, scored fairly, the default commuter setting hardly changed routes, because a time-proportional penalty caps the cost of a hazardous edge at $1 + \lambda s$. Larger risk weights reduced believed exposure smoothly, and at small detours the soft penalty reduced expected exposure more than

hard blocking did. Calibration against a proxy label was weak and did not favour class-specific decay. Next steps are a time-stamped corpus with independent ground truth from the 2023 floods and the coming monsoon, a harm term that does not scale with traversal time, a Wilson-type bound [60], [61] in place of (3), discounting of correlated reports, use of traversal without reports as negative evidence, and measurement of verifier pass rates on a low-cost phone.

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